Sequences and Series

Sequence limits

1 Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$

b $a_n = \frac{n^2}{2^n}$

c $a_n = (-1)^n$

2 Evaluate $\lim_{n\to\infty}\left(1+\frac{3}{n}\right)^n$.

Series and partial sums

3 Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

4 Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

5 Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

Monotonic and bounded sequences

6 Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n.

7 Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.

More series practice

8 Evaluate the series $\sum_{n=1}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right]$ by splitting into two geometric series.

9 A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.

Recursive sequences

10 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2, a_3, and a_4.

11 Assuming the recursive sequence above converges to a limit L, find L by setting $L = \frac{L+6}{2}$ and solving.

Repeating decimals as series

12 Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.